

Debug With Shubham Notes

TCS NQT Test Curriculum 2025

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TCS NQT Foundation Aptitude Curriculum

Number System

Mensuration

Arithmetic Ability

Elementary Statistics

Data Interpretation

Simplification & Approximation

Number System

1- Divisibility

2- Number System , HCF & LCF

3- Numbers and Decimal Frations

Decimal

In mathematics, a decimal is a number that is expressed in the base-10 number system, using the digits 0 to 9. It is called "decimal" because it is based on powers of 10.

Fraction

A fraction is a way of expressing a part of a whole or a ratio between two quantities. It represents a division of one quantity (the numerator) by another quantity (the denominator).

Number, Decimal and Fraction Formulas

$$(a + b)(a - b) = (a^2 - b^2)$$

$$2. (a + b)^2 = (a^2 + b^2 + 2ab)$$

$$3. (a - b)^2 = (a^2 + b^2 - 2ab)$$

$$4. (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$5. (a^3 + b^3) = (a + b)(a^2 - ab + b^2)$$

$$6. (a^3 - b^3) = (a - b)(a^2 + ab + b^2)$$

$$7. (a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$$

$$8. \text{ when } a + b + c = 0, \text{ then } a^3 + b^3 + c^3 = 3abc$$

Question 1. Find a number such that when 18 is subtracted from 8 times the number, the result is 12 more than twice the number.

- A. 5
- B. 10
- C. 15
- D. 20

Let, the number be z , Then,

$$8z - 18 = 2z + 12$$

$$\Rightarrow 6z = 30 \Leftrightarrow z = 5.$$

Hence, the required number is 5.

Question 2. The sum of two numbers is 198. If one-third of the one exceeds one - seventh of the other by 8, find the smaller number.

- A. 65
- B. 68
- C. 70.8
- D. 76.2

Let the number be z and $(198 - z)$. then,

$$\Rightarrow z/3 - (198 - z)/7 = 8.$$

$$\Rightarrow 7z - 3(198 - z) = 168$$

$$\Rightarrow 10z = 762$$

$$\Rightarrow z = 76.2$$

Question 3. What percent of 1000.008 is 10320.408?

- A. 1130.25%
- B. 1032.032%
- C. 1030%
- D. 1032%

Let $x\%$ of 1000.008 = 10320.408

Then, $x/100 * 1000.008 = 10320.408$

$$1000.008x = 1032040.8$$

$$x = 1032040.8 / 1000.008$$

$$x = 1032.032\%$$

Question 4. Find the value of x in the given expression: $310420/1009 = x/0.0075$

A. 1.8

B. 2.30

C. 3.5

D. None of the above

$$\begin{aligned} 310420/1009 &= x/0.0075 \\ &= 1009x = 310420 * 0.0075 \\ &= 1009x = 2328.15 \\ x &= 2328.15/1009 \\ x &= 2.30 \end{aligned}$$

Question 5. If $122.21x = 0.00122y$, then find the value of $((y - x) / (y + x))$.

A. 6110439 / 6110561

B. 6110439/6110561

C. 6110561/6110500

None of the aboveD.

$$\begin{aligned} x/y &= 0.00122 / 122.21 \\ &= 000.122 / 12221 \\ &= 122 / 12221000 \\ &= 61 / 6110500 \\ x &= 61, y = 6110500 \\ y-x &= 6110500 - 61 = 6110439 \\ y+x &= 6110500+61 = 6110561 \\ (y-x) / (y+x) &= 6110439 / 6110561 \end{aligned}$$

Question 6. What least number is there in the place of * so that the number 9041705*4 is divisible by 7?

A. 3

B. 2

C. 4

D. 5

$$\begin{aligned} 9 + 0 + 4 + 1 + 7 + 0 + 5 + 4 &= 30 \\ \text{The nearest number divisible by 7 is } &35 \\ 7 * 5 &= 35 \\ \text{Therefore, the missing number is } &35 - 30 = 5 \\ \text{Hence, the number is } &904170554 \end{aligned}$$

Questions 7. Find the sum of all the prime number between 10000 and 10100?

A. 110424

B. 110648

C. 110651

D. 110650

The prime numbers between 10000 and 11000 are:

10007, 10009, 10037, 10039, 10061, 10067, 10069, 10079, 10091, 10093, and 10099

Therefore, the sum = $10007 + 10009 + 10037 + 10039 + 10061 + 10067 + 10069 + 10079 + 10091 + 10093 + 10099 = 110651$

Questions 8. A number is as much greater than 5001 as it is less than 10001. Find the number.

A. 7741

B. 7501

C. 7546

D. 7549

Let the number be x

Given, $(x - 5001) = (10001 - x)$

$= x + x = 10001 + 5001$

$2x = 15002$

$x = 7501$

Questions 9. Eleven thirtieth of a number is 12 more than one fifth of a number. Find the number.

A. 45

B. 60

C. 55

D. 72

$\frac{11}{30}x - \frac{1}{5}x = 12$

$\frac{5}{30}x = 12$

$\frac{1}{6}x = 12$

$x = 12 * 6$

$x = 72$

Questions 10. Find the value of $.114 * .114 + 2 * .114 * .044 + .044 * .044$.

A. 0.024964

B. 0.201232

C. 1.306956

D. 30.69

To calculate the expression $.114 * .114 + 2 * .114 * .044 + .044 * .044$, we can simplify it step by step:

$$.114 * .114 = 0.012996$$

$$2 * .114 * .044 = 0.010032$$

$$.044 * .044 = 0.001936$$

Adding these values together:

$$0.012996 + 0.010032 + 0.001936 = 0.024964$$

Therefore, the value of $.114 * .114 + 2 * .114 * .044 + .044 * .044$ is approximately 0.024964

Question 1 : $4100 + 13.952 - ? = 3764.002$

- A. 747.095
- B. 247.752
- C. 347.932
- D. 349.95

Explanation:

Let $4100 + 13.952 - x = 3764.002$.

Then $x = (4100 + 13.952) - 3764.002$

$= 4113.952 - 3764.002 = 349.95$.

Question 2 : What is the sum of the decimal fractions $25/100$ and $30/100$?

- A. $55/100$
- B. $65/100$
- C. $75/100$
- D. $85/100$

Solution:

Given decimal fractions: $25/100$ and $30/100$.

As the denominators are the same in both decimal fractions, we can directly add the numerators.

Thus,

$$(25/100) + (30/100) = (25 + 30)/100$$

$$(25/100) + (30/100) = 55/100.$$

Hence, the sum of the decimal fractions $25/100$ and $30/100$ is $55/100$.

Question 3 : 25% of 30% of $850 + 5 \times 76 = ?$

- A. 441.75
- B. 443.75
- C. 441.57
- D. 443.57

25% of 30% of $850 + 5 \times 76$

$$= 25/100 * 30/100 * 850 + 5 * 76$$

$$= 443.75$$

Question 4 : $(32/100) \times 750 - ? = (14/100) \times 540$

- A. 152.6
- B. 132.6
- C. 146.6
- D. 164.4

$$(32/100) \times 750 - ? = (14/100) \times 540$$

$$240 - x = 75.6$$

$$x = 240 - 75.6$$

$$x = 164.4$$

Question 5 : $\sqrt{196} \times \sqrt{144} \times 20\% \text{ of } 700 = ? + 1265$

- A. 22255
- B. 22266
- C. 22588
- D. 25874

$$\sqrt{196} \times \sqrt{144} \times 20\% \text{ of } 700 = ? + 1265$$

$$14 \times 12 \times 140 = ? + 1265$$

$$168 \times 140 = ? + 1265$$

$$23520 = ? + 1265$$

$$? = 23520 - 1265$$

$$? = 22255$$

Real Question and Ans

Question 1: What will be smallest fraction form of 0.0056

- A. $7/1275$
- B. $7/1250$
- C. $7/1175$
- D. $7/1150$

Question 2: Evaluate $\frac{(2.39^2 - 1.61^2)}{(2.39 - 1.61)}$

- A. 2
- B. 3
- C. 4
- D. 5

Question 3: If $144/0.144 = 14.4/x$, then $x = ?$

- A. 0.144
- B. 0.0144
- C. 0.00144
- D. 1.44

Question 4: What decimal of an hour is a second

- A. .0028
- B. .00027
- C. .0026
- D. .0025

Question 5: Arrange the fractions in ascending order

- A. $\frac{5}{8} < \frac{7}{12} < \frac{3}{4} < \frac{13}{16}$
- B. $\frac{5}{8} < \frac{7}{12} < \frac{13}{16} < \frac{3}{4}$
- C. $\frac{5}{8} < \frac{3}{4} < \frac{13}{16} < \frac{7}{12}$
- D. $\frac{7}{12} < \frac{5}{8} < \frac{3}{4} < \frac{13}{16}$

Question 6: $789.009 - ? + 56.84 = 215.943$

- A. 899.015
- B. 752.804
- C. 714.642
- D. 629.906

Question 7: $12^{4.5} \times 12^? = 12^{5.6} \times 12^{1.8}$

- A. 1.5
- B. 1.8
- C. 2.9
- D. 3.1

Question 8: 1% of $\{(9.99 \times 9.9 \times 9) \div 9.9 \div 0.3\} = ?$

- A. 2.76
- B. 2.39
- C. 3.10
- D. 2.99

Question 9: $0.3 \times 10^{-3} = ?$

- A. 0.0003
- B. 0.003
- C. 0.03
- D. 0.3

Question 10: $(1.1 + 1.1 + 1.1 + 1.1) \times 1.1 \times 1.1 = ? \times 0.121$

- A. 54
- B. 33
- C. 44
- D. 22

Solution

Ans -1

First, we need to convert 0.0056 to a fraction.

We can do this by moving the decimal point to the right until we get a whole number, and then putting this over the appropriate power of 10:

$$0.0056 = 56/10000$$

Now we need to simplify this fraction. To do this, we'll divide both the numerator and denominator by their greatest common divisor (GCD).

The factors of 56 are: 1, 2, 4, 7, 8, 14, 28, 56

The factors of 10000 are: 1, 2, 4, 5, 8, 10, 16, 20, 25, 40, 50, 100, 125, 200, 250, 500, 1000, 2000, 2500, 5000, 10000

The greatest common divisor is 8.

$$56 \div 8 = 7$$

$$10000 \div 8 = 1250$$

Therefore, the simplified fraction is $7/1250$

Ans 2

First, let's calculate the numerator ($2.39^2 - 1.61^2$):

$$2.39^2 = 5.7121$$

$$1.61^2 = 2.5921$$

$$5.7121 - 2.5921 = 3.12$$

Now, let's calculate the denominator ($2.39 - 1.61$):

$$2.39 - 1.61 = 0.78$$

So our expression now looks like:

$$3.12 / 0.78$$

Dividing these:

$$3.12 \div 0.78 = 4$$

Therefore, $(2.39^2 - 1.61^2) / (2.39 - 1.61) = 4$

Ans -3

We start with the equation: $144/0.144 = 14.4/x$

First, let's simplify the left side of the equation:

$$144/0.144 = 144 \times (1/0.144) = 144 \times 10/1.44 = 1000$$

Now our equation looks like this:

$$1000 = 14.4/x$$

To solve for x, we can multiply both sides by x:

$$1000x = 14.4$$

Now divide both sides by 1000:

$$x = 14.4/1000 = 0.0144$$

Check our answer choices:

The correct answer is B. 0.0144

Ans-4

First, let's recall how many seconds are in an hour:
 $60 \text{ minutes} \times 60 \text{ seconds} = 3600 \text{ seconds in an hour}$

Now, we need to express 1 second as a fraction of an hour:

$$1 \text{ second} = 1/3600 \text{ of an hour}$$

To convert this fraction to a decimal, we divide 1 by 3600:

$$1 \div 3600 = 0.0002777... \text{ (repeating)}$$

Looking at our options, we need to round this to 4 decimal places.

0.0002777... rounds to 0.0003 (4 decimal places)

However, none of our options have 0.0003. The closest option is 0.00027.

Therefore, the correct answer is B. .00027

Ans - 5

$$5/8 = 0.625$$

$$7/12 = 0.5833 \text{ (rounded to 4 decimal places)}$$

$$3/4 = 0.75$$

$$13/16 = 0.8125$$

Now, let's arrange these decimals in ascending order:

$$0.5833 < 0.625 < 0.75 < 0.8125$$

Converting back to fractions, we get:

$$7/12 < 5/8 < 3/4 < 13/16$$

This matches option D.

Ans - 6

We have the equation: $789.009 - ? + 56.84 = 215.943$

Let's call the unknown number x . So our equation becomes:

$$789.009 - x + 56.84 = 215.943$$

To isolate x , first let's subtract 56.84 from both sides:

$$789.009 - x = 215.943 - 56.84$$

$$789.009 - x = 159.103$$

Now, let's subtract 789.009 from both sides:

$$-x = 159.103 - 789.009$$

$$-x = -629.906$$

Finally, let's multiply both sides by -1 :

$$x = 629.906$$

Ans - 7

We have the equation: $12^{4.5} * 12^? = 12^{5.6} * 12^{1.8}$

In exponential equations with the same base, we can add the exponents. So let's add the exponents on each side:

Left side: $4.5 + ?$

Right side: $5.6 + 1.8 = 7.4$

Now our equation looks like this:

$$12^{(4.5 + ?)} = 12^{7.4}$$

For this to be true, the exponents must be equal:

$$4.5 + ? = 7.4$$

To solve for ?, let's subtract 4.5 from both sides:

$$? = 7.4 - 4.5 = 2.9$$

Ans -8

First, let's calculate what's inside the curly brackets {}.
 $(9.99 \times 9.9 \times 9) \div 9.9 \div 0.3$

Simplify the multiplication first:

$$9.99 \times 9.9 \times 9 = 889.11$$

Now our expression looks like:

$$889.11 \div 9.9 \div 0.3$$

Divide by 9.9:

$$889.11 \div 9.9 = 89.81$$

Now divide by 0.3:

$$89.81 \div 0.3 = 299.37$$

So the value inside the curly brackets is 299.37

Now we need to find 1% of 299.37:

$$1\% \text{ of } 299.37 = 299.37 \times 0.01 = 2.9937$$

Rounding to two decimal places:

$$2.9937 \approx 2.99$$

Ans - 9

0.3×10^{-3} means we're multiplying 0.3 by 0.001 (which is what 10^{-3} equals).

$$0.3 \times 0.001 = 0.0003$$

Alternatively, we can think of it as moving the decimal point in 0.3 three places to the left (because the exponent is -3):
 $0.3 \rightarrow 0.03 \rightarrow 0.003 \rightarrow 0.0003$

Therefore, $0.3 \times 10^{-3} = 0.0003$, which matches option A.

Ans - 10

First, let's calculate the left side of the equation:
 $(1.1 + 1.1 + 1.1 + 1.1) \times 1.1 \times 1.1$

Simplify the addition inside the parentheses:
 $(4.4) \times 1.1 \times 1.1$

Multiply:

$$4.4 \times 1.1 = 4.84$$

$$4.84 \times 1.1 = 5.324$$

So, the left side equals 5.324

Now, our equation looks like this:
 $5.324 = ? \times 0.121$

To solve for ?, we divide both sides by 0.121:
 $5.324 \div 0.121 = ?$

Calculate:

$$5.324 \div 0.121 = 44$$

Therefore, the correct answer is C. 44.